Better GDM

The Problem

The idea of GDM, modelling matrices of dissimilarity, was suggested by Ferrier (2002). Recently, White et al. (2023) suggested improvements to generalized dissimilarity modelling. Here we will develop an alternative, based on modelling the data rather than the distances between data.

The ecological problem is to explain how beta diversity, i.e. variation between sites, is affected by predictors, such as the environment. This can be done using GDM, which regresses distances between communities against teh environemntal covariates. Here we suggest an alternative approach.

The Model

We assume we have a site by species matrix, Y, with each row being observations of species (abundance, presentece etc.), and each column being a site. So Y_{ij} is the observation for the i^th species ($i = 1, \dots, S$) and j^th species ($j = 1, \dots, N$). For each site we have P covariates, x_{jk} ($k = 1, \dots, P$), so these are in a $N \times P$ matrix X.

We assume we model this data using a GLM, i.e. we assume Y_{ij} follows some distribution in the exponential family, with expected value $\mu = g(\eta)$, where $g()$ is a link function, and η is a linear predictor:

$$\eta_{ij} = \alpha_i + \kappa_j + \beta_{ij}$$

For identifiability we assume $\sum_j \kappa_j = 0$ and $\sum_j \beta_j = 0$

So α_i is the mean commonness of a species, and κ is the abundance at a site: we will assume this is not interesting, as it is affected by the sampling process. β_{ij} is where beta diversity lies (at least compositional beta diversity). The total beta diversity can be measured as $\mathrm{Var}(\beta) = \sigma^2_\beta$. We can clearly model β_{ij} further, e.g. for a single covariate x_j

$$\beta_{ij} = \beta_0 + \delta_i x_j + \epsilon_{ij}$$

i.e. each species has its own response to x.

Then,

$$\begin{aligned}
\mathrm{Var}(\beta) &= \mathrm{Var}(\beta_0 + \delta_i x_j + \epsilon_{ij}) \\
&= \frac{1}{(S-1)(N-1)} \sum_{i=1}^{S} \sum_{j=1}^{N} (\delta_i (x_j - \bar{x}) + \epsilon_{ij})^2 \\
&= \frac{1}{(S-1)(N-1)} \left(\sum_{i=1}^{S} \sum_{j=1}^{N} \delta_i^2 (x_j - \bar{x})^2 + \sum_{i=1}^{S} \sum_{j=1}^{N} \epsilon_{ij}^2 \right) \\
&= \frac{1}{(S-1)} \sum_{i=1}^{S} \delta_i^2 \frac{1}{(N-1)} \sum_{j=1}^{N} (x_j - \bar{x})^2 + \frac{1}{(S-1)(N-1)} \sum_{i=1}^{S} \sum_{j=1}^{N} \epsilon_{ij}^2 \\
&= \frac{1}{(S-1)} \sum_{i=1}^{S} \delta_i^2 \mathrm{Var}(x) + \mathrm{Var}(\epsilon_{ij}) \\
&= \frac{\mathrm{Var}(x)}{(S-1)} \sum_{i=1}^{S} (\delta_i - \bar{\delta} + \bar{\delta})^2 + \mathrm{Var}(\epsilon_{ij}) \\
&= \frac{\mathrm{Var}(x)}{(S-1)} \sum_{i=1}^{S} (\delta_i - \bar{\delta})^2 - S\bar{\delta}^2 + \mathrm{Var}(\epsilon_{ij}) \\
&= \mathrm{Var}(\delta_i)\mathrm{Var}(x_j) + \frac{S}{S-1} \bar{\delta}^2 \mathrm{Var}(x) + \mathrm{Var}(\epsilon_{ij})
\end{aligned}$$

because $\sum_i (\delta_i - \bar{\delta} + \bar{\delta})^2 = \sum_i (\delta_i - \bar{\delta})^2 + \sum_i (\delta_i - \bar{\delta})\bar{\delta} + \sum_i \bar{\delta})^2 = \sum_i (\delta_i - \bar{\delta})^2 + \bar{\delta}(\sum_i \delta_i - S\bar{\delta}) + S\bar{\delta}^2 = \sum_i (\delta_i - \bar{\delta})^2 + S\bar{\delta}^2$.

So, we can model the species abundance as a function of the environment. We can extend this to more than one covariate, if they are assumed to have independent effects between species (i.e. δ_{ik} is independent of δ_{il} for species i and covariates k and l).

Relationship to Distance

If we have two vectors of biodiversity, η_1 and η_2 then the distance between then the distance between them is just the Euclidean distance:

$$D_{12} = \|\eta_1 - \eta_2\| = \sqrt{\sum_{i=1}^{S} (\eta_{i1} - \eta_{i2})^2}$$

So that

$$
\begin{aligned}
D_{12}^2 &= \sum_{i=1}^{S} (\eta_{i1} - \eta_{i2})^2 \\
&= \sum_{i=1}^{S} \left(\alpha_i + \kappa_1 + \beta_{i1} - (\alpha_i + \kappa_2 + \beta_{i2})\right)^2 \\
&= \sum_{i=1}^{S} \left((\kappa_1 - \kappa_2) + (\beta_{i1} - \beta_{i2})\right)^2 \\
&= \sum_{i=1}^{S} \left((\kappa_1 - \kappa_2)^2 + (\kappa_1 - \kappa_2)(\beta_{i1} - \beta_{i2}) + (\beta_{i1} - \beta_{i2})^2\right) \\
&= S(\kappa_1 - \kappa_2)^2 + (\kappa_1 - \kappa_2) \sum_{i=1}^{S} (\beta_{i1} - \beta_{i2}) + \sum_{i=1}^{S} (\beta_{i1} - \beta_{i2})^2 \\
&= S(\kappa_1 - \kappa_2)^2 + \sum_{i=1}^{S} (\beta_{i1} - \beta_{i2})^2 \\
&= S(\kappa_1 - \kappa_2)^2 + \sum_{i=1}^{S} \beta_{i1}^2 - 2 \sum_{i=1}^{S} \beta_{i1}\beta_{i2} + \sum_{i=1}^{S} \beta_{i1}^2 \\
&= S(\kappa_1 - \kappa_2)^2 + (S-1)\left(\mathrm{Var}(\beta_{\cdot 1}) + \mathrm{Var}(\beta_{\cdot 2}) - 2\mathrm{Cov}(\beta_{\cdot 1}\beta_{\cdot 2})\right)
\end{aligned}
$$

Where the $(\kappa_1 - \kappa_2) \sum_{i=1}^{S} (\beta_{i1} - \beta_{i2})$ term disappears because, by construction, $\sum_{i=1}^{S} \beta_{ij} = 0$. $\mathrm{Var}_{ij}(X)$ is the variance of X in populations i and j.

If we are interested in compositional diversity, we can ignore the terms involving the κs, so the distance is just a function of the second moments of β.

Examples

We use the three examples from White et al. (2023). First we need some bookkeeping, to list the libraries and create a function to …

```
library(zen4R) # usedto download SA data
library(INLA)
library(gdm)
library(taxize)

# Define PC priors, to stop the models exploding horribly
PCHyper <- list(prec = list(prior="pc.prec", param=c(u=0.5, alpha=0.001)))
PCHyper2 <- list(prec = list(prior="pc.prec", param=c(u=1, alpha=0.1)))
```

```r
# Number of simulations from posterior to calculate statistics
nsim <- 1000

# Function to calculate variance components
CalcPostVar <- function(INLAsamp, data) {
  # Extract main effects of covariates
  LatentNames <- attr(INLAsamp$latent, "dimnames")[[1]]
  RFnames <- paste(c(gsub("Precision for ", "", names(INLAsamp$hyperpar)),
                     "(Intercept)", "Predictor"), collapse="|")
  FxNames <- LatentNames[!grepl(RFnames, LatentNames)]
  RFNames <- gsub(":.", "", FxNames)
  # Calculate/extract parts of variance terms
  VarHyper <- 1/INLAsamp$hyperpar
  wh.delta <- sapply(RFNames, function(wh, nm) grep(wh, nm), nm=names(VarHyper))
  VarDelta <- VarHyper[wh.delta]
  VarOth <- VarHyper[!seq_along(VarHyper)%in%wh.delta]

  VarX <- var(data[, RFNames])
  S <- nlevels(data$Species)

  # Deltabar
  Fx.wh <- which(attr(INLAsamp$latent, "dimnames")[[1]]%in%FxNames)
  Deltabar <- INLAsamp$latent[Fx.wh]
  names(Deltabar) <- attr(INLAsamp$latent, "dimnames")[[1]][Fx.wh]


  AllVars <- c(MeanDelta = t(Deltabar)%*%VarX%*%Deltabar,
               VarDelta = sum(VarDelta*diag(VarX)), # not sure this is right
               VarOth)
  return(AllVars)
}

# Function to transform log precision of hyper parameters to standard deviation
CalcHyperSDs <- function(mod, use=NULL) {
  if(is.null(use)) use <- seq_along(mod$internal.marginals.hyperpar)

  sd.lst <- lapply(mod$internal.marginals.hyperpar[use], function(lst) {
    marg <- inla.tmarginal(function(x) exp(-x/2), lst)
    res <- c(inla.mmarginal(marg), inla.hpdmarginal(c(0.5, 0.95), marg))
    names(res) <- c("mode", "Lower50", "Lower95", "Upper50", "Upper95")
    res
  })
```

```
  sd.df <- as.data.frame(do.call(rbind, sd.lst))
  sd.df
}

# Function to create linear combinations of fixed & random effects for variable var
CreateLinComb <- function(var, Levels, prefix="Species.") {
  lc <- inla.make.lincombs(var= rep(1, nlevels(Levels)),
                           tt = diag(1, nlevels(Levels)))
  lc <- lapply(lc, function(l, var) {
    names(l[[1]]) <- var
    names(l[[2]]) <- paste0(prefix, var)
    l
  }, var=var)
  names(lc) <- paste0(var, names(lc))
  lc
}


# Function to plot fixed effects
# PlotFixed(fixed=SAFamSumm$fixed, use=Use.fixed)

PlotFixed <- function(fixed, use) {
  plot(fixed[use, "mean"], use, yaxt="n", xlab="Mean Effect", ylab="",
       xlim=range(fixed[use, c("0.025quant", "0.975quant")]),
       main="Mean Effect")
  segments(fixed[use, "0.025quant"], use, fixed[use, "0.975quant"], use)
  axis(2, gsub("\\.sc", "", rownames(fixed)[use]), at=use, las=1)
  abline(v=0, lty=2)
}

# Function to plot hyperparameter SDs
# PlotHyper(SDs=SAHyperSDs, use=Use.hyper)
PlotHyper=function(SDs, use=NULL) {
  if(is.null(use)) use <- 1:nrow(SDs)
  plot(SDs[use,"mode"], use, yaxt="n", xlab="Std Dev", ylab="",
       xlim=range(c(0,SDs[,c("Lower95", "Upper95")])),
       main="Std Dev across Species")
  segments(SDs[use,"Lower50"], use, SDs[use,"Upper50"], use, lwd=3, lend=2)
  segments(SDs[use,"Lower95"], use, SDs[use,"Upper95"], use)
}

# Function to plot random effects and their means
```

```
PlotEff <- function(summ, mean, ...) {
  At <- rank(summ$mean)
  plot(summ$mean, At, yaxt="n", type="n", ...)
  rect(mean["0.025quant"], -10, mean["0.975quant"], 1e4, col="pink", border=NA)
  abline(v=mean["mean"], col=2)
  points(summ$mean, At)
  segments(summ[,"0.025quant"], At,
           summ[,"0.975quant"], At)
  abline(v=0, lty=2)
}
```

Example 1: South African flora

This is data with 288 species and 413 sites. The data is given in wide format, so we need to convert it to long. The data can be downloaded from Zenodo.

```
#|output: false

# Download the data, if it hasn't already been done. Cleans up afterwards, too
if(!file.exists("data/sa_species_data.csv")) {
  download_zenodo("10.5281/zenodo.10091441", path="data")
  unzip("data/spGDMM-code-spGDMM_v1.zip", exdir="data")
  DataFiles <- dir("data/philawhite-spGDMM-code-d8279ee/data/")
  sapply(DataFiles, function(x) {
    cat(x, file.exists(file.path("data/philawhite-spGDMM-code-d8279ee/data/",x)), "\n")
    file.copy(from=file.path("data/philawhite-spGDMM-code-d8279ee/data/",x),
              to=file.path("data/",x), overwrite = TRUE)
  })
    unlink("data/philawhite-spGDMM-code-d8279ee/", recursive = TRUE)
    unlink("data/spGDMM-code-spGDMM_v1.zip", recursive = TRUE)
}

Data <- read.csv("data/sa_species_data.csv")


# SpNames <- gsub("\\.", " ", names(Data)[grep("\\.", names(Data))])
SpNames <- names(Data)[grep("\\.", names(Data))]
# FamNames <- names(FamData)[!names(FamData)%in%names(Data)]

# Discover that not all family names from GBIF are he same as the family data
# FamData <- read.csv("philawhite-spGDMM-code-d8279ee/data/sa_family_data.csv")
```

```r
# FamNames <- names(FamData)[!names(FamData)%in%names(Data)]
# Family[!Family%in%FamNames]

SpData.wide <- Data[, SpNames]
IsZero <- apply(SpData.wide, 2, function(x) mean(x==0))

UseNames <- names(IsZero)[IsZero<0.99]
RemoveNames <- SpNames[!SpNames%in%UseNames]
NamesToLong <- names(Data)[!names(Data)%in%RemoveNames]

SAData.long <- reshape(Data[,NamesToLong], direction = "long", varying = UseNames,
                       v.names = "Percent", timevar="SpeciesID")
SAData.long$Species <- SpNames[SAData.long$SpeciesID]
SAData.long$Percent2 <- pmax(0.1, SAData.long$Percent)
SAData.long$Abund <- log(SAData.long$Percent2/(100-SAData.long$Percent2))

SA.EnvCov <- c("gmap", "RFL_CONC", "HeatLoadIndex30m", "tmean13c",
            # "Elevation30m",
            "SoilConductivitymSm", "SoilTotalNPercent")

ScCov <- apply(SAData.long[,SA.EnvCov], 2, scale)
colnames(ScCov) <- paste0(colnames(ScCov), ".sc")

SAData <- cbind(SAData.long, ScCov)

# Get family names from GBIF
# Classification <- classification(SpNames, db = 'gbif', rows=1)
# Family <- unlist(lapply(Classification, function(lst) lst$name[lst$rank=="family"]))
#
# SAData$Family <- unlist(sapply(SAData$Species, function(Sp, fam) {
# #  cat(Spp, "\n")
#   ff <- which(Sp==names(fam))
#   if(length(ff)==0) {
#     res <- "AAAAAGH"
#   } else {
#     res <- fam[ff]
#   }
#   res
# }, fam=Family))
#
# # any(SAData$Family=="AAAAAGH")
#
```

```r
# SAData$Family <- factor(SAData$Family)

# SAData$Species <- factor(SAData$Species)
SAData$plot <- factor(SAData$plot)
```

Covariates:

- gmap: mean annual precipitation
- RFL_CONC: rainfall concentration
- Elevation30m: elevation (at 30m resolution)
- HeatLoadIndex30m: heat load index (at 30m resolution)
- tmean13c: mean annual temperature
- SoilConductivitymSm: soil conductivity
- SoilTotalNPercent: Nitrogen concentration in soil (%)

We remove elevation because of the high correlation with tmean13c (-0.94), and all species that occur in <1% of sites, giving 146 species out of the original 288.

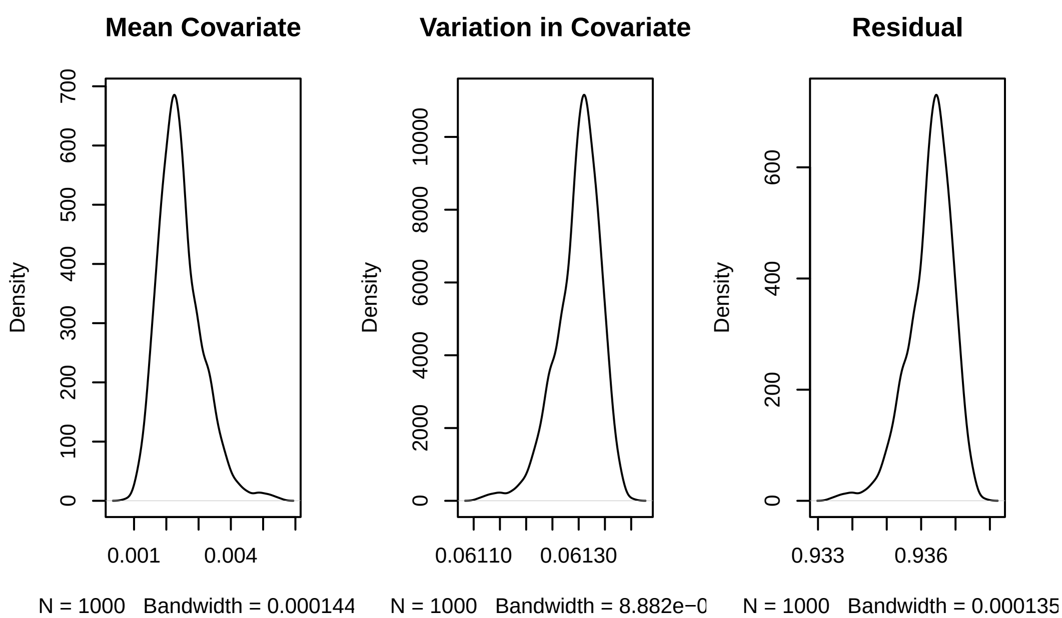
We fit the model with INLA, and use relatively tight PC priors to control the estimates of the environmental effects (without doing this everything is horribly unstable).

```r
# Create formula for INLA
EnvForm <- sapply(SA.EnvCov, function(var) {
  paste0(var, ".sc + f(Species.", var, ".sc, ", var,
         ".sc, hyper = PCHyper)")
})

SA.Formula <- formula(
  paste0("Abund ~ f(Species, hyper = PCHyper) + f(plot, hyper = PCHyper) + ",
         paste(EnvForm, collapse=" + "))
)

SAenvInd <- data.frame(sapply(SA.EnvCov, function(nm, dat)
  assign(nm, dat), dat=SAData$Species))
names(SAenvInd) <- paste0("Species.", names(SAenvInd), ".sc")

SAData <- cbind(SAData, SAenvInd)

LCs.lst <- lapply(paste0(SA.EnvCov, ".sc"), CreateLinComb,
                  Levels=factor(SAData$Species))
SA.LCs <- do.call(c, LCs.lst)

# int.strategy = "ccd" or int.strategy = "grid"
```

```r
# Fit the model
#.  Note: this is not terribly stable
SAinla <- inla(SA.Formula, data=SAData, control.compute=list(config = TRUE),
               control.inla=list(int.strategy = "ccd"), lincomb = SA.LCs)
SASumm <-summary(SAinla)
```

This will calculate the statistics

```r
SAsamps <- inla.posterior.sample(nsim, SAinla)

SAPosts <- data.frame(t(sapply(SAsamps, CalcPostVar, data=SAData)))
Use <- !names(SAPosts)%in%paste0("Precision.for.", c("Species", "plot"))
SA.PropVar <- sweep(SAPosts[,Use], 1, rowSums(SAPosts[,Use]), "/")
names(SA.PropVar) <- gsub("Precision.for.the.Gaussian.observations", "VarEps",
                          names(SA.PropVar), fixed=TRUE)

par(mfrow=c(1,3))
plot(density(SA.PropVar$MeanDelta), main="Mean Covariate")
plot(density(SA.PropVar$VarDelta), main="Variation in Covariate")
plot(density(SA.PropVar$VarEps), main="Residual")
```



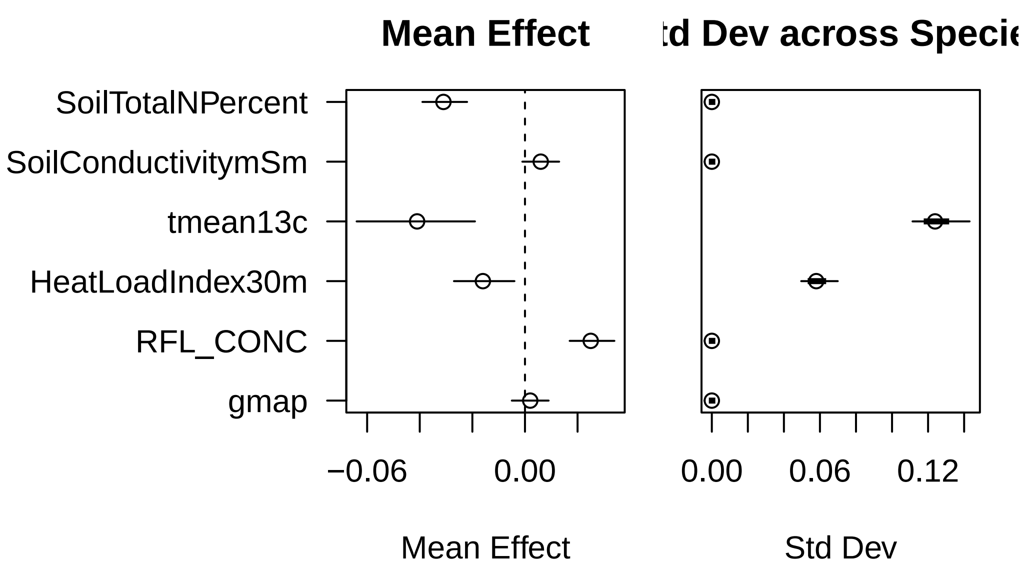

```r
# Summary
SA.SummVars <- cbind(
```

```r
  Mode=MCMCglmm::posterior.mode(coda::as.mcmc(SA.PropVar)),
  coda::HPDinterval(coda::as.mcmc(SA.PropVar), prob = 0.95)
)
```

We find that 3.67% of the variation in beta diversity is explained by the covariates, with a 95% HPDI of 3.58% - 3.77%.

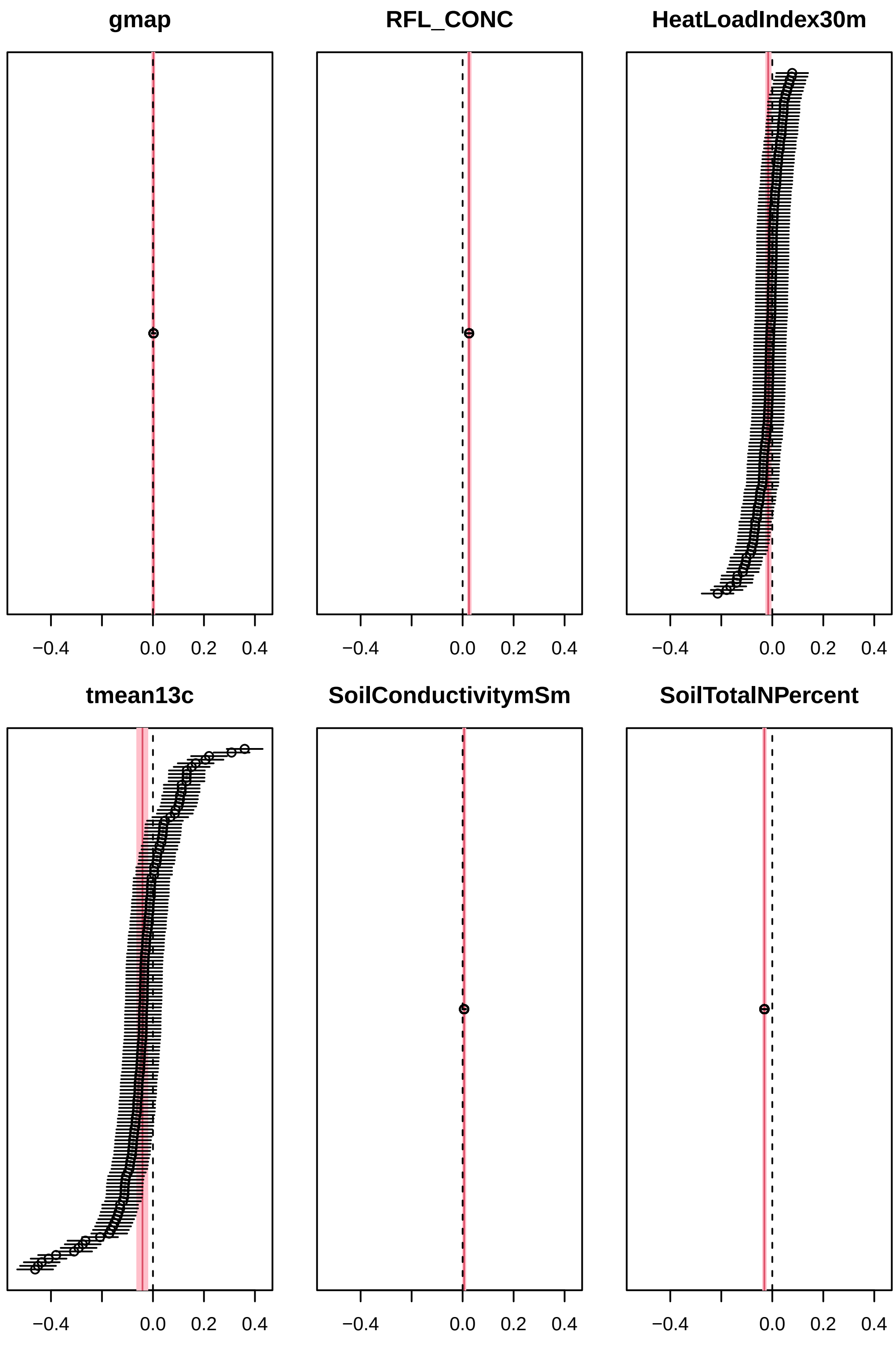
We can look at the importance of the variables. First, by plotting the mean effects and their standard deviations (across species). Then we can look at their contributions to the variance.

```r
# Plot means
Use.fixed <- which(!grepl("\\(Int", rownames(SASumm$fixed)))
Use.hyper <- sapply(SA.EnvCov, function(x, y) grep(x, y),
                    y=names(SAinla$marginals.hyperpar))

SAHyperSDs <- CalcHyperSDs(SAinla, use=Use.hyper)

par(mar=c(4.1,1,3,1), oma=c(1,8,1,1), mfrow=c(1,2))
 PlotFixed(fixed=SASumm$fixed, use=Use.fixed)
 PlotHyper(SDs=SAHyperSDs, use=NULL)
```



We can see that temperature has the largest effect, both in the average effect and the variation across species

```
SA.RanEff <- SAinla$summary.lincomb.derived
SA.UseSumRan <- grep("Species\\.", names(SAinla$summary.random))
Xlims <- range(SA.RanEff[,c("0.025quant", "0.975quant")])

par(mar=c(2.1,1,3,1), mfrow=c(2,3))
sapply(SA.EnvCov, function(env, RanEff, SummFix, ...) {
PlotEff(summ=RanEff[grep(env, rownames(RanEff)),],
        mean=SummFix[paste0(env, ".sc"),],
        ylab="", xlab="", main=env, ...)
}, RanEff=SA.RanEff, SummFix=SASumm$fixed, xlim=Xlims)
```


gmap
RFL_CONC
HeatLoadIndex30m
tmean13c
SoilConductivitymSm
SoilTotalNPercent
−0.4
0.0
0.2
0.4

```
$gmap
NULL

$RFL_CONC
NULL

$HeatLoadIndex30m
NULL

$tmean13c
NULL

$SoilConductivitymSm
NULL

$SoilTotalNPercent
NULL
```

Example 1a: South African flora at the family level

White et al. (2023) looked at the family level for this data, not the species. Soi we will too. This means we have 52 families and 413 sites.

```
#|output: false

FamData <- read.csv("data/sa_family_data.csv")
FamNames <- names(FamData)[!names(FamData)%in%names(Data)]
FamData.wide <- FamData[, FamNames]
IsZero <- apply(FamData.wide, 2, function(x) mean(x==0))

UseNames <- names(IsZero)[IsZero<0.99]
RemoveNames <- FamNames[!FamNames%in%UseNames]
NamesToLong <- names(FamData)[!names(FamData)%in%RemoveNames]

FamData.long <- reshape(FamData[,NamesToLong], direction = "long",
                        varying = UseNames,
                       v.names = "Percent", timevar="FamilyID")

FamData.long$Species <- FamNames[FamData.long$FamilyID]
FamData.long$Percent2 <- pmax(0.1, FamData.long$Percent)
FamData.long$Abund <- log(FamData.long$Percent2/(100-FamData.long$Percent2))
```

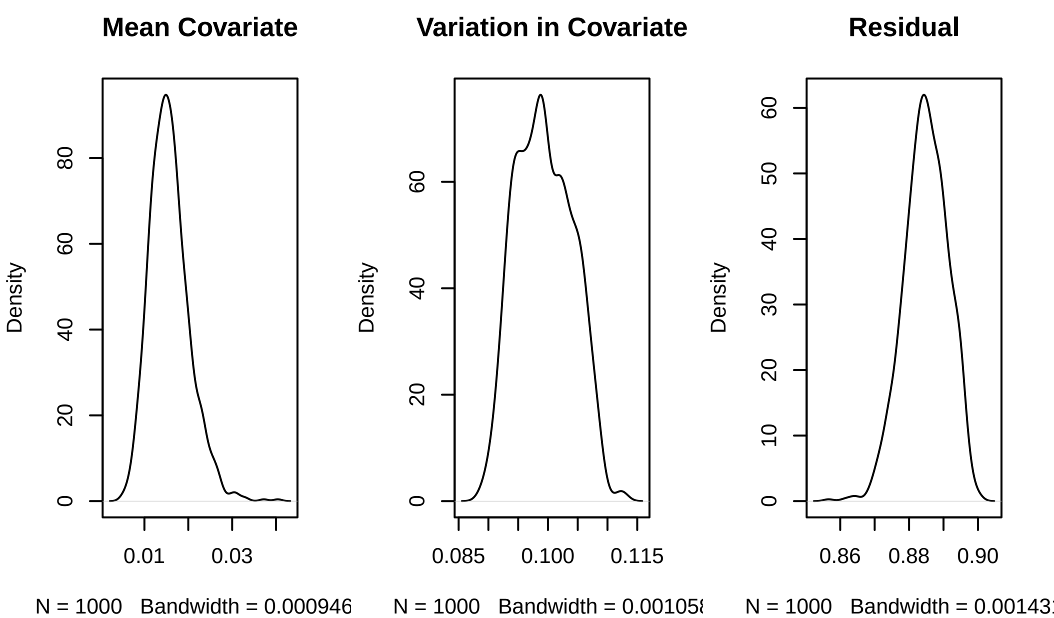
```r
Fam.EnvCov <- c("gmap", "RFL_CONC", "HeatLoadIndex30m", "tmean13c",
            # "Elevation30m",
            "SoilConductivitymSm", "SoilTotalNPercent")

ScCov <- apply(FamData.long[,Fam.EnvCov], 2, scale)
colnames(ScCov) <- paste0(colnames(ScCov), ".sc")

SAFamData <- cbind(FamData.long, ScCov)

SAFamData$Family <- factor(UseNames[SAFamData$FamilyID])
SAFamData$plot <- factor(SAFamData$plot)
```

The covariates are as before, as is the use of INLA and the priors

```r
# Create formula for INLA
SAFam.Formula <- formula(
  paste0("Abund ~ f(Family, hyper = PCHyper) + f(plot, hyper = PCHyper) + ",
         paste(
           c(paste0(Fam.EnvCov, ".sc"),
             sapply(paste0(Fam.EnvCov, ".sc"), function(eu) {
               paste0("f(Family.", eu, ", ", eu, ", hyper = PCHyper)")
             })), collapse=" + "))
)

SAFamenvInd <- data.frame(sapply(Fam.EnvCov, function(nm, dat) assign(nm, dat),
                              dat=SAFamData$Family))
names(SAFamenvInd) <- paste0("Family.", names(SAFamenvInd), ".sc")
SAFamData <- cbind(SAFamData, SAFamenvInd)

# Linear combinations
LCs.lst <- lapply(paste0(SA.EnvCov, ".sc"), CreateLinComb,
                  Levels=factor(SAFamData$Family), prefix="Family.")
SAFam.LCs <- do.call(c, LCs.lst)

# Fit the model
SAFaminla <- inla(SAFam.Formula, data=SAFamData,
                  control.compute=list(config = TRUE),
               control.inla=list(int.strategy = "ccd"), lincomb = SAFam.LCs)
SAFamSumm <-summary(SAFaminla)
```

This will calculate the statistics

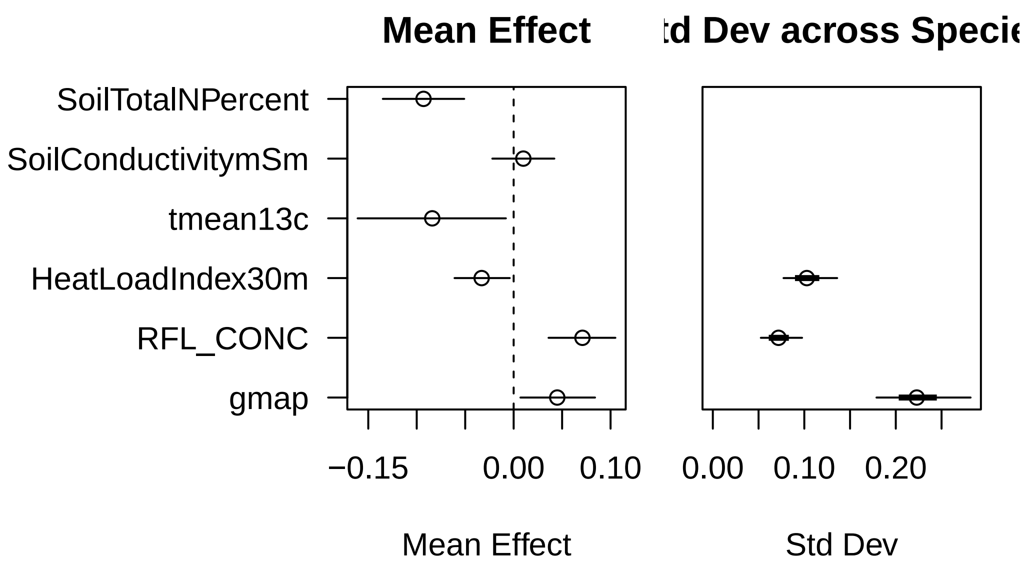
```r
SAFamsamps <- inla.posterior.sample(nsim, SAFaminla)

SAFamPosts <- data.frame(t(sapply(SAFamsamps, CalcPostVar, data=SAFamData)))
Use <- !names(SAFamPosts)%in%paste0("Precision.for.", c("Family", "plot"))
SAFam.PropVar <- sweep(SAFamPosts[,Use], 1, rowSums(SAFamPosts[,Use]), "/")
names(SAFam.PropVar) <- gsub("Precision.for.the.Gaussian.observations", "VarEps",
                             names(SAFam.PropVar), fixed=TRUE)

par(mfrow=c(1,3))
plot(density(SAFam.PropVar$MeanDelta), main="Mean Covariate")
plot(density(SAFam.PropVar$VarDelta), main="Variation in Covariate")
plot(density(SAFam.PropVar$VarEps), main="Residual")
```




```r
# Summary
SAFam.SummVars <- cbind(
  Mode=MCMCglmm::posterior.mode(coda::as.mcmc(SAFam.PropVar)),
  coda::HPDinterval(coda::as.mcmc(SAFam.PropVar), prob = 0.95)
)
```

We find that 11.7% of the variation in beta diversity is explained by the covariates, with a 95% HPDI of 10.38% - 12.72%.

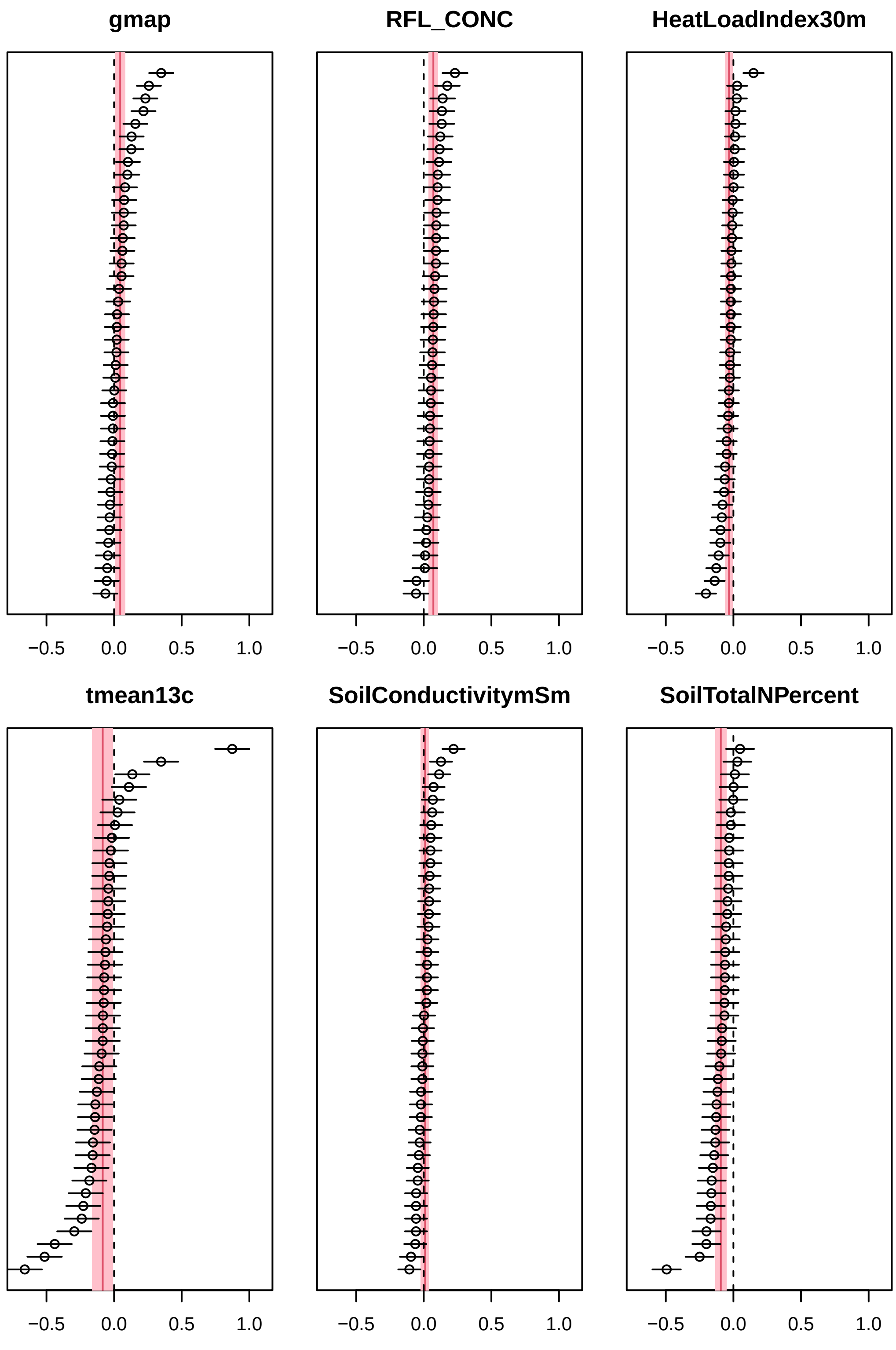
```r
Use.fixed <- which(!grepl("\\(Int", rownames(SAFamSumm$fixed)))
Use.hyper <- sapply(SA.EnvCov, function(x, y) grep(x, y),
                    y=names(SAFaminla$marginals.hyperpar))

SAFamHyperSDs <- CalcHyperSDs(SAFaminla, use=Use.hyper)

par(mar=c(4.1,1,3,1), oma=c(1,8,1,1), mfrow=c(1,2))
 PlotFixed(fixed=SAFamSumm$fixed, use=Use.fixed)
 PlotHyper(SDs=SAFamHyperSDs, use=Use.hyper)
```



We can see that temperature has the largest effect, both in the average effect and the variation across species

```r
SA.RanEff <- SAinla$summary.lincomb.derived
SA.UseSumRan <- grep("Species\\.", names(SAinla$summary.random))
Xlims <- range(SA.RanEff[,c("0.025quant", "0.975quant")])

SAFam.RanEff <- SAFaminla$summary.lincomb.derived
UseSumRan <- grep("Family\\.", names(SAFaminla$summary.random))
Xlims <- range(unlist(lapply(SAFaminla$summary.random[UseSumRan], function(lst)
  range(lst[,c("0.025quant", "0.975quant")]))))


par(mar=c(2.1,1,3,1), mfrow=c(2,3))
sapply(SA.EnvCov, function(env, RanEff, SummFix, ...) {
PlotEff(summ=RanEff[grep(env, rownames(RanEff)),],
```

```
        mean=SummFix[paste0(env, ".sc"),],
        ylab="", xlab="", main=env, ...)
}, RanEff=SAFam.RanEff, SummFix=SAFamSumm$fixed, xlim=Xlims)
```


gmap
RFL_CONC
HeatLoadIndex30m
tmean13c
SoilConductivitymSm
SoilTotalNPercent
−0.5
0.0
0.5
1.0

```
$gmap
NULL

$RFL_CONC
NULL

$HeatLoadIndex30m
NULL

$tmean13c
NULL

$SoilConductivitymSm
NULL

$SoilTotalNPercent
NULL
```

Comparison with spGDMM

With spGDMM, White et al. (2023) could compare models using cross-validation, were able to compare the effects of the different covariates, and were qualitatively able to compare the covariate effects with a spatial term:

```
The scale of the differences in $\eta(\mathbf{s}, \mathbf{s}')$ is large relative to the sc
```

Here we don't (yet) have a spatial term, but we can compare the covaraite effects to the total variance in diversity, and see that at teh species level almost all of the variation (88.3%) is unexplained, whilst at the family level about 11.7% of teh variation is explained by the covaraitesx

Example 2: The Inevitable BCI Data

This is counts of trees on an island that only exists to let ecologists take a holiday in the jungle. The data can be downloaded from Condit et al. (2002).

```r
BCI.Env <- read.csv("data/Panama_env.csv")
BCI.Sp <- read.csv("data/Panama_species.csv")

# Stupid naming
# "species" is actually site. And the format of the names is different between files
```

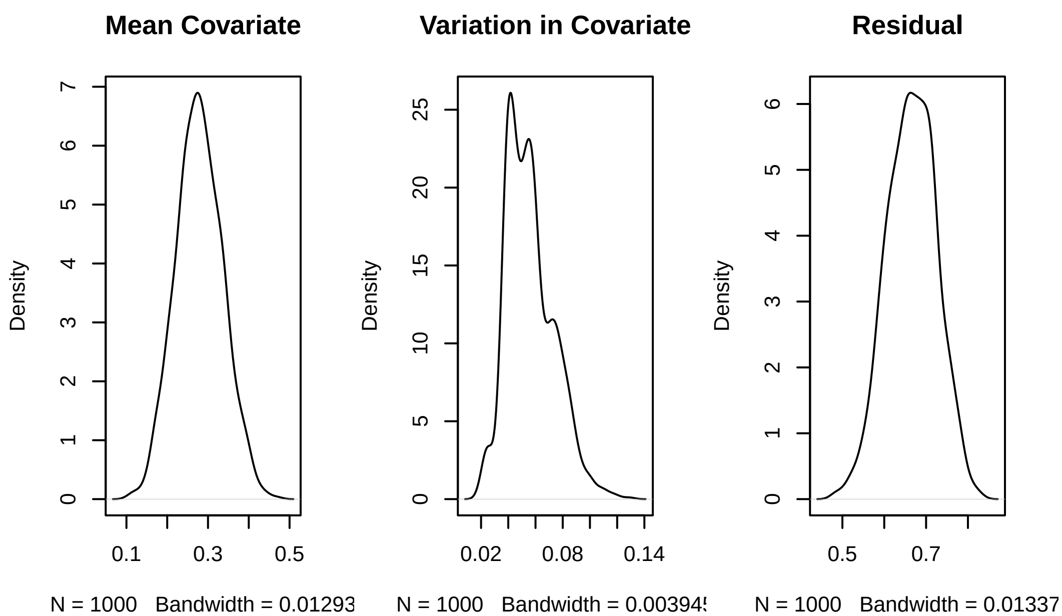
```r
# This is why "data scientist" has a lower job satisfaction than "statistician"
BCI.Sp$site.no. <- sprintf("P%02d", as.numeric(gsub("P", "", BCI.Sp$species)))
BCI.Sp$species <- NULL

BCI.EnvCov <- c("precip", "elev", "age")
Pres <- apply(BCI.Sp, 2, function(x) sum(x>0))
sp.tmp <- BCI.Sp[,which(Pres>10)]
dat.tmp <- cbind(BCI.Env, sp.tmp)
SpNames <- names(sp.tmp)[!names(sp.tmp)%in%c("species", "site.no.")]
BCIData <- reshape(dat.tmp, direction = "long", varying = SpNames, v.names = "Count")
BCIData$Species <- factor(SpNames[BCIData$time])
BCIData$Site <- factor(BCIData$site.no.)
BCIData$SiteSpecies <- paste0(BCIData$Site, BCIData$Species)
# Scale env
Scale.tmp <- apply(BCIData[,BCI.EnvCov], 2, scale)
colnames(Scale.tmp) <- paste0(colnames(Scale.tmp), ".sc")
BCIData <- cbind(BCIData, Scale.tmp)
```

We can fit the model in INLA:

```r
BCI.Formula <- formula(
  paste0("Count ~ f(Species, hyper = PCHyper) + f(Site, hyper = PCHyper) + ",
         paste(
           c(paste0(BCI.EnvCov, ".sc"),
             sapply(paste0(BCI.EnvCov, ".sc"), function(eu) {
               paste0("f(Species.", eu, ", ", eu, ", hyper = PCHyper)")
             })), collapse=" + "), " + f(OD, hyper = PCHyper)")
)

BCIenvInd <- data.frame(sapply(BCI.EnvCov, function(nm, dat) assign(nm, dat),
                               dat=BCIData$Species))
names(BCIenvInd) <- paste0("Species.", names(BCIenvInd), ".sc")

BCIData <- cbind(BCIData, BCIenvInd)
BCIData$OD <- factor(1:nrow(BCIData))

# Linear combinations
# Should improve code
LCs.lst <- lapply(paste0(BCI.EnvCov, ".sc"), CreateLinComb, Levels=BCIData$Species)
BCI.LCs <- do.call(c, LCs.lst)

# Fit the model
```

```r
BCIinla <- inla(BCI.Formula, data=BCIData, control.compute=list(config = TRUE),
                family="poisson", lincomb = BCI.LCs)
BCISumm <-summary(BCIinla)
```

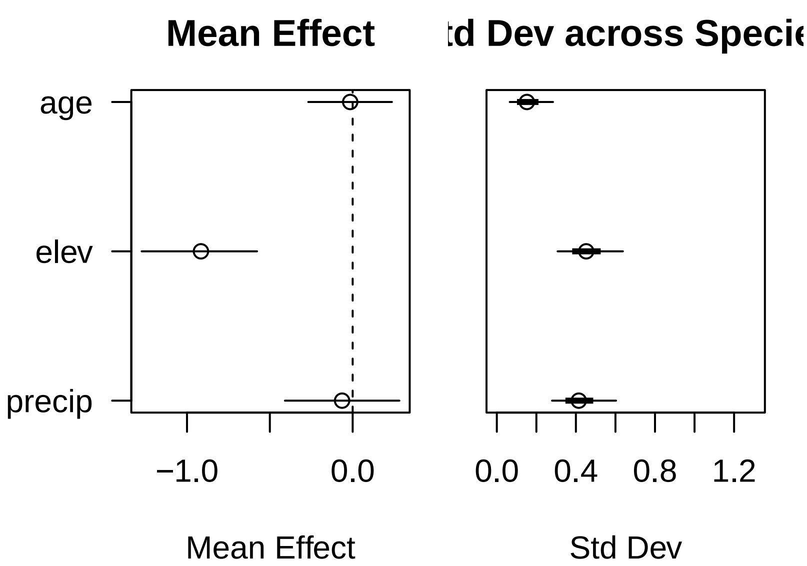
```r
BCIsamps <- inla.posterior.sample(nsim, BCIinla)

BCI.Posts <- data.frame(t(sapply(BCIsamps, CalcPostVar, data=BCIData)))
Use <- !names(BCI.Posts)%in%paste0("Precision.for.", c("Species", "Site"))
BCI.PropVar <- sweep(BCI.Posts[,Use], 1, rowSums(BCI.Posts[,Use]), "/")
names(BCI.PropVar)[3] <- "VarEps"

par(mfrow=c(1,3))
plot(density(BCI.PropVar$MeanDelta), main="Mean Covariate")
plot(density(BCI.PropVar$VarDelta), main="Variation in Covariate")
plot(density(BCI.PropVar$VarEps), main="Residual")
```




```r
BCI.SummVars <- cbind(
  Mode=MCMCglmm::posterior.mode(coda::as.mcmc(BCI.PropVar)),
  coda::HPDinterval(coda::as.mcmc(BCI.PropVar), prob = 0.95)
)
```

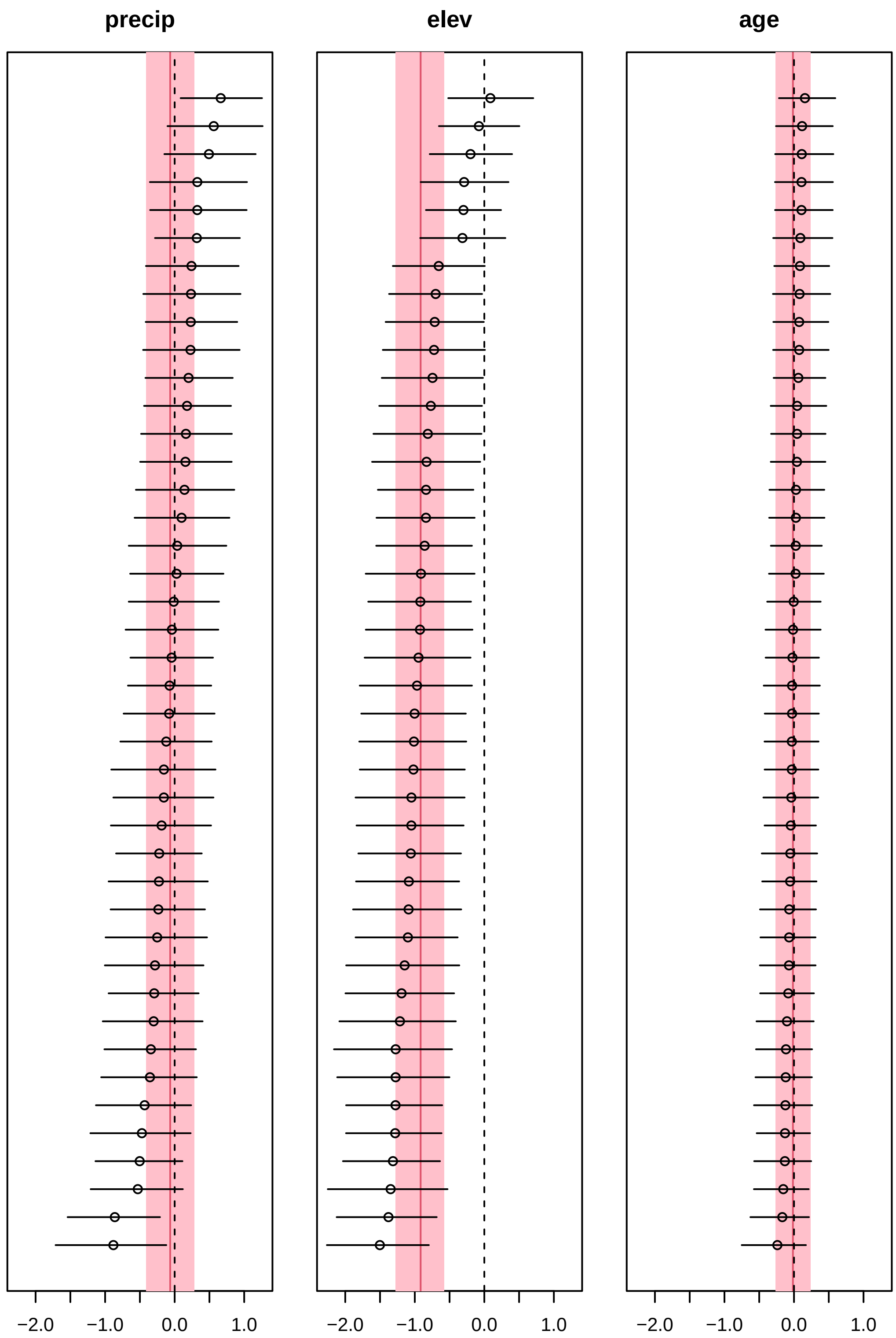
We find that 29.63% of the variation in beta diversity is explained by the covariates, with a 95% HPDI of 21.59% - 44.13%.

```r
# Plot means
Use.fixed <- which(!grepl("\\(Int", rownames(BCISumm$fixed)))
Use.hyper <- sapply(BCI.EnvCov, function(x, y) grep(x, y),
                    y=names(BCIinla$marginals.hyperpar))

BCIHyperSDs <- CalcHyperSDs(BCIinla)

par(mar=c(4.1,1,3,1), oma=c(1,8,1,1), mfrow=c(1,2))
 PlotFixed(fixed=BCISumm$fixed, use=Use.fixed)
 PlotHyper(SDs=BCIHyperSDs, use=Use.hyper)
```



We can see that elevation has the largest average effect, but there is variation across species in all effects

```r
BCI.RanEff <- BCIinla$summary.lincomb.derived

UseSumRan <- grep("Species\\.", names(BCIinla$summary.random))
Xlims <- range(BCI.RanEff[,c("0.025quant", "0.975quant")])

par(mar=c(2.1,1,3,1), mfrow=c(1,3))
sapply(BCI.EnvCov, function(env, RanEff, Fixed, ...) {
  PlotEff(summ=RanEff[grep(env, rownames(RanEff)),],
        mean=Fixed[grep(env, rownames(Fixed)),],
        ylab="", xlab="", main=env, ...)
}, RanEff=BCI.RanEff, Fixed=BCISumm$fixed, xlim=Xlims)
```


precip
elev
age
−2.0
−1.0
0.0
1.0

```
$precip
NULL

$elev
NULL

$age
NULL
```

Example 3: Australian Plants

This is occurrence of plants in Southwest Australia, the `southwest` data in the `gdm` package. The data are not well documented, with several presences per species per site. Absence data is not given. Fortunately, the environmental data does not vary within a site.

```
data("southwest")

SW.EnvCov <- names(southwest)[!names(southwest)%in%c("species","site", "Lat", "Long")]

Pres <- table(southwest$species)
sp.keep <- names(Pres[Pres>50]) #reduce to 10?

southwest.small <- southwest[southwest$species%in%sp.keep,]
southwest.small$species <- factor(southwest.small$species)
SW.pres <- expand.grid(species=unique(southwest.small$species),
                       site=unique(southwest.small$site))
SW.pres$Presence <- apply(SW.pres, 1, function(wh, dat) {
 any(dat$site==wh["site"] & dat$species==wh["species"])
}, dat=southwest.small)

# Extract environmental variables
SWenv <- data.frame(apply(southwest.small[,SW.EnvCov], 2, function(x, dat) {
  tapply(x, list(dat$site), function(xx) xx[1])
}, dat=southwest.small))
env.site <- rownames(SWenv)
SWenv <- data.frame(apply(SWenv, 2, scale))
names(SWenv) <- paste0(colnames(SWenv), ".sc")
SWenv$site <- env.site

SW.data <- merge(SW.pres, SWenv)
SW.data$Presence <- as.numeric(SW.data$Presence)
```

```r
# Check to see if there is variation in environment between sites. There isn't.
# SDs <- by(southwest.small, list(southwest.small$site), function(df, vars) {
#   apply(df[,vars], 2, sd)
# }, vars=SW.EnvCov)
# table(unlist(SDs))
```

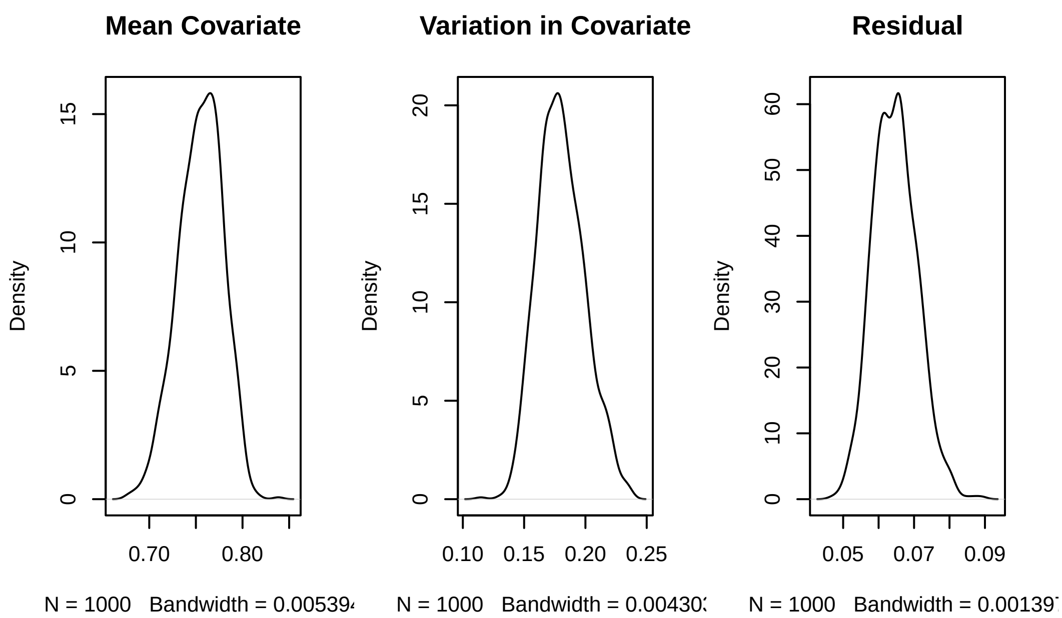
We can fit the model in INLA:

```r
SW.Formula <- formula(
  paste0("Presence ~ f(species, hyper = PCHyper) + f(site, hyper = PCHyper) + ",
         paste(
           c(paste0(SW.EnvCov, ".sc"),
             sapply(paste0(SW.EnvCov, ".sc"), function(eu) {
               paste0("f(Species.", eu, ", ", eu, ", hyper = PCHyper)")
             })), collapse=" + "))
)

SWenvInd <- data.frame(sapply(SW.EnvCov, function(nm, dat) assign(nm, dat),
                              dat=SW.data$species))
names(SWenvInd) <- paste0("Species.", names(SWenvInd), ".sc")
SW.data <- cbind(SW.data, SWenvInd)

# Create linear combinations
LCs.lst <- lapply(paste0(SW.EnvCov, ".sc"), CreateLinComb, Levels=SW.data$species)
SW.LCs <- do.call(c, LCs.lst)

# Fit the model
SWinla <- inla(SW.Formula, data=SW.data, control.compute=list(config = TRUE),
                family="binomial", Ntrials = 1, lincomb = SW.LCs)
SWSumm <-summary(SWinla)
```

And now…

```r
SWsamps <- inla.posterior.sample(nsim, SWinla)

SW.Posts <- data.frame(t(sapply(SWsamps, CalcPostVar, data=SW.data)))
SW.Posts$logitVar <- pi/3
Use <- !names(SW.Posts)%in%paste0("Precision.for.", c("species", "site"))
SW.PropVar <- sweep(SW.Posts[,Use], 1, rowSums(SW.Posts[,Use]), "/")
names(SW.PropVar)[3] <- "VarEps"
```

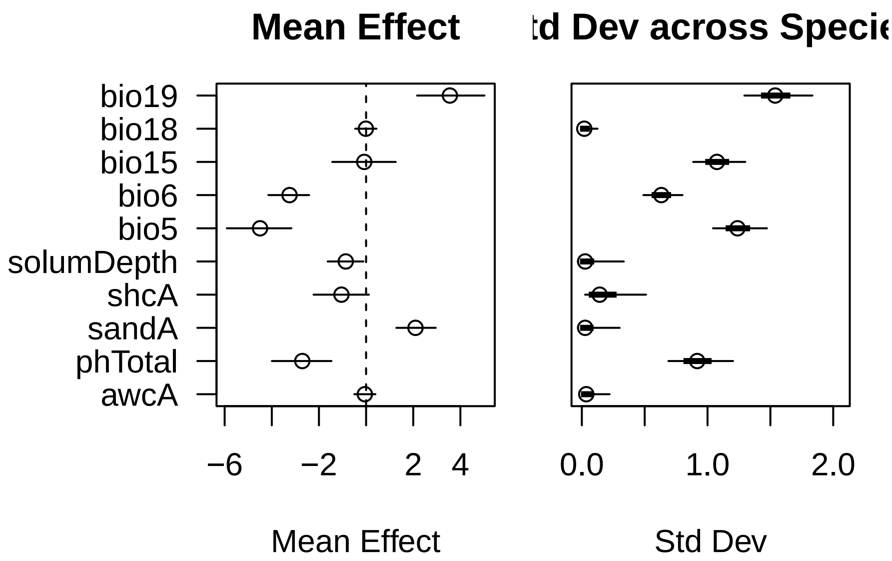
```r
par(mfrow=c(1,3))
plot(density(SW.PropVar$MeanDelta), main="Mean Covariate")
plot(density(SW.PropVar$VarDelta), main="Variation in Covariate")
plot(density(SW.PropVar$VarEps), main="Residual")
```




```r
SW.SummVars <- cbind(
  Mode=MCMCglmm::posterior.mode(coda::as.mcmc(SW.PropVar)),
  coda::HPDinterval(coda::as.mcmc(SW.PropVar), prob = 0.95)
)
```

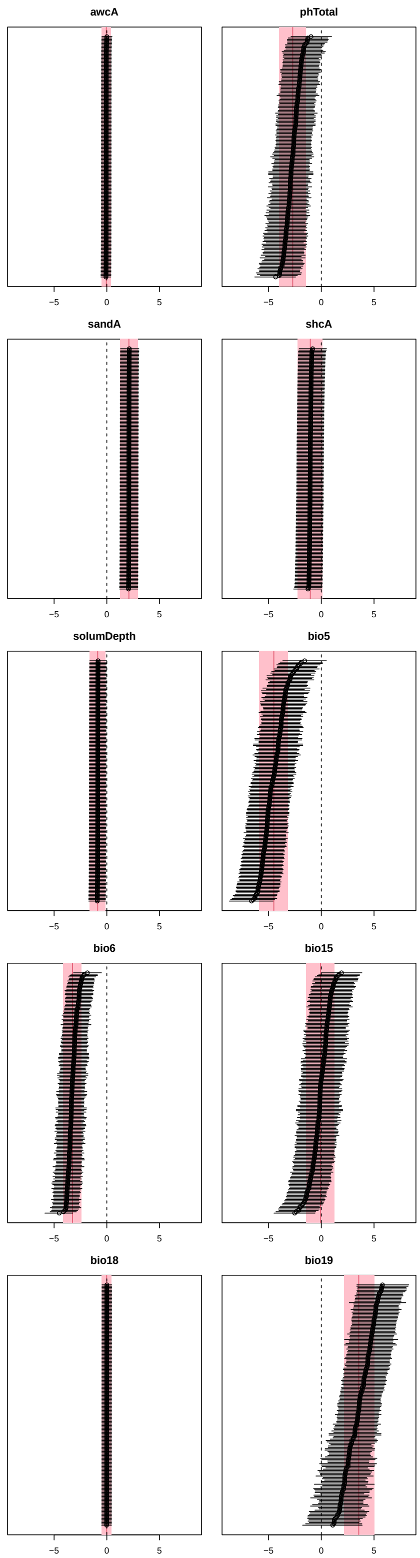
We find that 93.4% of the variation in beta diversity is explained by the covariates, with a 95% HPDI of 92.4% - 94.77%.

```r
# Plot means
Use.fixed <- which(!grepl("\\(Int", rownames(SWSumm$fixed)))
Use.hyper <- sapply(SW.EnvCov, function(x, y) grep(x, y),
                    y=names(SWinla$marginals.hyperpar))

SWHyperSDs <- CalcHyperSDs(SWinla)

par(mar=c(4.1,1,3,1), oma=c(1,8,1,1), mfrow=c(1,2))
 PlotFixed(fixed=SWSumm$fixed, use=Use.fixed)
 PlotHyper(SDs=SWHyperSDs, use=Use.hyper)
```



We can see that elevation has the largest average effect, but there is variation across species in all effects

```
SW.RanEff <- SWinla$summary.lincomb.derived

UseSumRan <- grep("Species\\.", names(SWinla$summary.random))
Xlims <- range(SW.RanEff[,c("0.025quant", "0.975quant")])

par(mar=c(2.1,1,3,1), mfrow=c(5,2))

sapply(SW.EnvCov, function(env, RanEff, Fixed, ...) {
  PlotEff(summ=RanEff[grep(env, rownames(RanEff)),],
        mean=Fixed[grep(env, rownames(Fixed)),],
        ylab="", xlab="", main=env, ...)
}, RanEff=SW.RanEff, Fixed=SWSumm$fixed, xlim=Xlims)
```


awcA
phTotal
sandA
shcA
solumDepth
bio5
bio6
bio15
bio18
bio19
−5
0
5

```
$awcA
NULL

$phTotal
NULL

$sandA
NULL

$shcA
NULL

$solumDepth
NULL

$bio5
NULL

$bio6
NULL

$bio15
NULL

$bio18
NULL

$bio19
NULL
```

References

Condit, Richard, Nigel Pitman, Egbert G. Leigh, Jérôme Chave, John Terborgh, Robin B. Foster, Percy Núñez, et al. 2002. "Beta-Diversity in Tropical Forest Trees." *Science* 295 (5555): 666–69. https://doi.org/10.1126/science.1066854.

Ferrier, Simon. 2002. "Mapping Spatial Pattern in Biodiversity for Regional Conservation Planning: Where to from Here?" *Systematic Biology* 51 (2): 331–63. https://doi.org/10.1080/10635150252899806.

White, Philip A., Henry A. Frye, Jasper A. Slingsby, John A. Silander Jr, and Alan E. Gelfand. 2023. "Generative Spatial Generalized Dissimilarity Mixed Modelling (spGDMM): An Enhanced Approach to Modelling Beta Diversity." *Methods in Ecology and Evolution* n/a (n/a). https://doi.org/https://doi.org/10.1111/2041-210X.14259.